

EB-H-JEPA: Energy-Based Hamiltonian JEPA World Models

A Failure Atlas and a Controlled Sample-Efficiency Test in Crafter

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Abstract

World models that operate in a learned latent space (JEPA; LeCun 2022) are the leading bet for sample-efficient, long-horizon model-based RL (DreamerV3, Hafner et al. 2023). A recurring hypothesis is that imposing physical structure on the latent predictor — Hamiltonian, metriplectic, or GENERIC — should make learned dynamics faithful and stable. We test this hypothesis twice, with instruments of increasing fidelity. First, on a nonlinearly lifted Lorenz-63 system, we audit five structurally distinct latent predictors under a chaos-aware protocol. The result is a failure atlas: each constraint class fails, but on a property disjoint from the one it enforces. A rigid Hamiltonian predictor freezes and cannot dissipate; a naive metriplectic predictor sits on a dead $LL^\top$ saddle with dissipation pinned at exactly zero; fixing both bugs (constant divergence-free J , nonzero R initialization) makes dissipation engage and recovers the true phase-space contraction -13.667 to $\sim 1\%$ accuracy — yet spectral alignment still cannot pin individual Lyapunov exponents. Second, and constructively, we insert the fixed-metriplectic predictor into a compact DreamerV3-style agent and measure sample efficiency on Crafter (Hafner 2021), the benchmark of the world-model group that introduced RSSM. If structure helps, this is the first positive evidence that physical inductive biases pay off in an RL world model; if it hurts, that is the honest negative the field needs. All code, runs, and evidence are public; every claim traces to a JSON artifact.

1 Introduction

JEPA-style world models [12, 2, 5] and their model-based RL instantiations [9, 10, 3] share a core bet: a latent predictor learned from observations can model dynamics that stay stable and meaningful over long horizons. The predictor is the crux. DreamerV3’s categorical RSSM learns it from data alone; a growing line of work asks whether injecting physical structure — symplectic/Hamiltonian form [6, 4, 16], metriplectic/GENERIC dissipation [15, 17, 11], or learned divergence gates [7] — yields predictors that are more faithful and more sample-efficient.

This paper reports a two-part test of that hypothesis, built to be reproducible end to end.

Part 1: a failure atlas on a known chaotic system. We train five structurally distinct latent predictors on Lorenz-63 lifted into a 64-D latent chart and audit each under a chaos-aware protocol: rollout boundedness, leading Lyapunov exponent, phase-space contraction, attractor overlap, and dissipation engagement. The honest headline is negative, and more precise than “structure doesn’t help”: *every constraint class fails, but on a property disjoint from the one it enforces*. A rigid Hamiltonian predictor (fixed canonical J , learned H) is conservative by construction and cannot dissipate; it freezes into a quasi-periodic tube with near-zero motion. A naive metriplectic predictor learns R through $R = LL^\top$ but sits on the dead saddle where $L = 0$: R stays exactly 0, dissipation never engages. Fixing both bugs — constant divergence-free J (closing the learned-skew loophole) and nonzero L initialization (breaking the dead saddle) — makes R engage and recovers the true contraction -13.667 to $\sim 1\%$. Yet even the best arm cannot pin individual Lyapunov exponents: differentiable QR spectral shaping controls only a finite-horizon statistic, not the asymptotic Oseledets spectrum [14]. And overlap metrics reward dead models: a frozen predictor sits inside the encoded attractor and posts the best Chamfer score while being dynamically inert.

Part 2: a controlled sample-efficiency test in Crafter. A failure atlas on Lorenz tells us where structure fails in isolation; it does not tell us whether structure helps or hurts an *actual RL world model*, where the predictor co-adapts with an actor trained by latent imagination. We build a compact but faithful reimplement of the DreamerV3 recipe — CNN encoder/decoder, recurrent memory, categorical posterior, KL free bits, symlog targets, TD(λ) imagination — in which the stochastic prior is pluggable. The baseline is DreamerV3’s exact categorical RSSM; the treatment is a continuous latent whose prior mean advances under the fixed-metriplectic map from Part 1. Everything else is identical. We measure sample efficiency on Crafter [8], the open-world benchmark introduced by the RSSM group itself. The result — whether structure helps, hurts, or is neutral — is a controlled answer to the field’s open question, and either way it is publishable. Preliminary CPU verification (Sec. 4.2) confirms both arms train and the pipeline is sound; the headline numbers (Table 2) come from the 1-hour T4 runs whose exact commands are shipped in the repo.

2 Background

JEPA. Joint-Embedding Predictive Architectures [12] predict in representation space, not pixel space: an encoder produces a latent z_t from observation x_t , and a predictor advances it. The EB-JEPA line [2, 5] shows strong representation learning without reconstruction.

DreamerV3. Hafner et al. [9] train a recurrent state-space model (RSSM) — categorical stochastic prior $p(z_t | h_t)$, posterior $q(z_t | h_t, x_t)$, GRU memory h — jointly with reconstruction, reward, and continue heads, then train an actor-critic by “imagining” rollouts in the learned latent. It is the sample-efficiency reference for visual control [8].

Metriplectic / GENERIC. Dissipative physical systems are not Hamiltonian; they follow the metriplectic (GENERIC) form $\dot{z} = J\nabla H - R\nabla S$ [15], with J antisymmetric (energy-conserving flow), R symmetric positive semidefinite (dissipation), and H, S the energy and entropy potentials. Learned versions [17, 11, 7] parameterize J and R ; the two

failure modes isolated in this paper — learned J exploiting the skew loophole, and $R = LL^\top$ sitting at the dead $L = 0$ saddle — are exactly the traps a naive parameterization falls into.

3 Method

3.1 Latent predictors (Part 1)

All arms share: a 64-D latent chart (nonlinear diffeomorphic lift of Lorenz-63), an MLP dynamics predictor, SIGReg anti-collapse regularization, and identical training budgets. They differ only in the algebraic class of the predictor:

B Unconstrained MLP $z' = f_\theta(z)$.

C Rigid Hamiltonian $z' = z + dt J \nabla_z H(z, h)$, fixed canonical J ; conservative, cannot dissipate.

D Naive metriplectic, learned J , $R = L(z)L(z)^\top$, L initialized at zero.

E *Fixed* metriplectic: constant divergence-free J ($\text{div}(J \nabla H) = 0$ exactly), L initialized at 0.05, learned H, S conditioned on GRU state h .

F E plus differentiable-QR spectral alignment on the rollout Jacobian.

Chaos-aware evaluation: rollout boundedness, motion ratio $m = \mathbb{E}|\Delta z|/\mathbb{E}|z|$, leading exponent λ_1 by the QR method, contraction $\text{tr}(R)/n$ vs. true divergence -13.667 , and Chamfer/Hausdorff overlap against the encoded attractor.

3.2 Pluggable predictor in a DreamerV3-style agent (Part 2)

The treatment predictor is the Part-1 E-arm embedded in the RSSM:

$$z' = z + dt \left(\underbrace{J \nabla_z H(z, h)}_{\text{flow}} - \underbrace{R(z, h) \nabla_z S(z, h)}_{\text{dissipation}} \right), \quad R = LL^\top \succeq 0, \quad (1)$$

with constant canonical J (dimension 64), L initialized at 0.05, dt learnable, and H, S MLPs conditioned on the GRU memory h . The prior is Gaussian with this mean and fixed variance, keeping the KL analytic with free bits. The baseline is DreamerV3’s categorical RSSM (32 classes $\times$ 32 codes). All other components — encoder, decoder, reward/continue heads, TD(λ) imagination, actor-critic — are byte-identical across arms; any difference is attributable to the predictor structure alone.

4 Experiments

4.1 Part 1: failure atlas (3 seeds, real numbers)

Table 1 summarizes 3-seed runs (full traces in **results/**, runner in **benchmarks/**).

arm	motion ratio	model λ_1	contraction	$\text{tr}(R)/n$
B unconstrained	—	explodes	—	—
C rigid Hamiltonian	~ 0	~ 0	≈ 0	—
D naive metriplectic	low	overshoot	wrong	0.00000
E fixed metriplectic	1.1–2.9	1.0–4.3	$-\mathbf{13.80} \pm \mathbf{0.17}$	2.3–6.2
F + spectral align	1.2–3.8	1.4–5.0	-13.25 ± 0.19	2.5–8.3

Table 1: Failure atlas on lifted Lorenz-63. True divergence -13.667 ; true $\lambda_1 \approx 0.9$. E recovers contraction to $\sim 1\%$, but no arm pins λ_1 ; the rigid arm posts the best *overlap* while being dynamically dead.

Reading. (i) The dissipation gate works: once J is constant and R escapes the dead saddle, learned contraction matches truth. (ii) Spectral shaping does not: pinning the *sum* of the spectrum cannot pin any individual Oseledets exponent. (iii) Overlap metrics reward dead models; they must be gated on rollout liveness.

4.2 Part 2: Crafter sample efficiency (controlled)

Protocol. Two arms (RSSM baseline, metriplectic treatment), identical seeds and budgets; 20k environment steps; world model trained on batches of 16×16 -frame sequences; evaluation every 5k steps by greedy rollout return. KL free bits use a matched *total* budget of 32 bits across arms (1.0 per categorical code, as in DreamerV3, vs. 0.5 per continuous dimension), so posterior-collapse slack cannot confound the comparison. Every 2k steps the agent logs a latent-stability audit: raw (uncapped) prior/posterior KL, prior-vs-posterior drift normalized by latent scale, latent boundedness, and — for the metriplectic arm — dissipation engagement $\text{tr}(R)/n$. Runs execute with a hard wall-clock budget on a T4 (exact commands: `experiments/crafter/train.py`, `kaggle/README.md`).

Hypotheses. H_+ : structure helps — the metriplectic arm needs fewer environment steps to reach a given return, because its prior mean is a better dynamics prior (less posterior collapse, longer useful imagination). H_- : structure hurts — the continuous latent and learned potentials slow the world model relative to the discrete categorical prior. Either outcome is a publishable, controlled answer.

Preliminary verification (CPU, fake env). Both arms train end-to-end from a clean checkout: reconstruction loss falls, KL free bits active, actor/critic losses finite, evaluation episodes complete, and the stability audit runs — the metriplectic arm’s dissipation engages ($\text{tr}(R)/n > 0$ and rising) inside the RL loop, mirroring the Lorenz atlas. This validates the pipeline; it is not evidence about the hypothesis. The headline comparison (Table 2) is produced by the shipped T4 runs.

5 Related work

Structure-inducing predictors: Hamiltonian neural networks [6, 4, 16], metriplectic/GENERIC learning [17, 11, 7], latent-world-model physics [1, 13]. JEPA and model-based RL: [12, 2, 5, 9, 10, 3]. This paper differs in two ways: it documents *which* structural failure mode

arm	return @ 10k steps	return @ 20k steps
RSSM (DreamerV3)	–	–
Metriplectic (E-arm)	–	–

Table 2: Crafter sample efficiency, to be filled from the T4 runs (see [kaggle/README.md](#) for the one-click protocol). Placeholders are deliberate: we do not ship numbers we have not run.

each class exhibits (the atlas), and it runs the first controlled predictor-ablation inside a DreamerV3-style agent on the benchmark (Crafter) of the group that introduced RSSM.

6 Conclusion

Physical structure in latent predictors is not a free lunch: each naive class fails on a specific, diagnosable property. The fixes that make dissipation genuinely engage (constant divergence-free J ; broken dead saddle) are concrete and validated. Whether those fixes pay off inside a real RL world model is the open question we answer with the Crafter experiment — shipped, reproducible, and bounded by an hour of compute. The code, runs, and drafts are public; we invite scrutiny of every number.

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